

Research Article

Disentangling functional spaces: a multidimensional framework for ecology and conservation

Aurèle Toussaint¹, Pablo A. Tedesco¹, Gaël Grenouillet^{1,2}, Liis Kasari-Toussaint³ and Sébastien Brosse¹

¹Centre de Recherche sur la Biodiversité et l'Environnement (CRBE), Université de Toulouse, Toulouse, France

²Institut Universitaire de France, Paris, France

³Institute of Ecology and Earth Sciences, University of Tartu, Tartu, Estonia

Correspondence: Aurèle Toussaint (aurele.toussaint@cnrs.fr)

Ecography

2026: e08578

doi: [10.1002/ecog.08578](https://doi.org/10.1002/ecog.08578)

Subject Editor: Robert Fletcher

Editor-in-Chief:

Christine N. Meynard

Accepted 14 July 2026

Trait-based ecology often relies on a single functional space, where diverse traits are combined into one multivariate framework. While convenient, this approach risks masking ecological strategies because variance is often dominated by a few traits, such as body size. Here, we propose a modular alternative: function-related trait spaces, each representing a specific ecological role such as locomotion, reproduction, or diet. Using global data for > 10 000 bird species as study case, we show that function-related trait spaces capture complementary ecological information that is obscured in the combined trait space. The combined space consistently underestimates functional distinctiveness, overlooking more than 800 species identified as unique in function-related trait spaces. We call for integrating function-related trait spaces into biodiversity assessments to improve the detection and protection of irreplaceable ecological roles.

Keywords: birds, conservation, functional diversity, trait-based approach, trait space



Introduction

Complementing species identities with trait-based approaches has increased the generality and predictive power of community ecology regarding species' adaptation and biotic environment (Naeem and Wright 2003, Hooper et al. 2005, Grime 2006). Trait-based approaches have hence become a central paradigm in biodiversity science, offering a functional lens to study community structure (Bruehlheide et al. 2018, Jarzyna et al. 2020) and evolutionary dynamics (Deline et al. 2018, Pigot et al. 2020), ecosystem functioning (Gagic et al. 2015, Cadotte 2017), and understand the interplay between extinction risk and the ecological role of species (Ackerly and Cornwell 2007, Violle et al. 2007, Swenson 2016, Westoby 2024, Kopf et al. 2025). In the context of global changes, trait-based approaches can provide information about the future changes in ecosystem functioning and refine the strategies for preserving biodiversity



www.ecography.org

© 2026 The Author(s). Ecography published by John Wiley & Sons Ltd on behalf of Nordic Society Oikos

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

and sustaining ecosystem functioning (Pollock et al. 2017, Díaz et al. 2019, Sala et al. 2021). A central tool in this paradigm is the construction of a multidimensional space, called functional space (Díaz et al. 2015, Sosiak and Barden 2020, Carmona et al. 2021a, Mouillot et al. 2021, Toussaint et al. 2021), which aggregates diverse traits (both in nature and number) into a unified multivariate framework (Pollock et al. 2017, Rüger et al. 2020, Sala et al. 2021).

A fundamental assumption of the functional space paradigm is that diverse ecological traits – irrespective of their biological nature – can be integrated into a combined multivariate space (Blonder et al. 2014, Carmona et al. 2021a, Toussaint et al. 2021), as defined by Violle et al. (2007). This assumption implies that traits measuring morphology, life-history strategies, and resource use are commensurable – that is, that their joint variation can be meaningfully summarized along shared axes of a single multivariate space. We argue that this assumption is rarely justified in practice.

Commensurability would require that a unit of variation in beak length carries the same ecological meaning as a unit of variation in clutch size or dietary specialization – an assumption with no biological justification. In practice, combining incommensurable traits into a single distance matrix implicitly assigns weights to each ecological dimension that reflect its statistical variance rather than its ecological relevance, systematically privileging size-related axes of variation over trophic or reproductive ones (Carmona et al. 2021a). Ecologically meaningful dimensions of variation, such as trophic specialization or reproductive strategy, are consequently compressed into secondary axes or diluted entirely, creating an illusion of functional completeness while obscuring key axes of ecological differentiation.

This oversimplification operates through at least three mechanisms. First, traits belonging to different functional domains are shaped by distinct selective pressures and evolutionary constraints that operate at different timescales and phylogenetic levels (Deline et al. 2018, Pigot et al. 2020). Morphological traits reflect biomechanical and habitat-related constraints that evolve relatively slowly and are strongly conserved within lineages, whereas dietary traits can shift rapidly in response to resource availability and competitive interactions. Treating these dimensions as equivalent contributions to a shared ecological signal obscures the fact that they capture fundamentally different aspects of how species interact with their environment. Second, the axes of a combined functional space are mathematical constructs optimizing variance explanation across all traits simultaneously, not ecological axes reflecting defined biological processes. A species that is functionally extreme in one dimension, for instance a highly specialized nectarivore, may appear unremarkable in a combined space if its morphology and life-history are average, because its trophic distinctiveness is diluted by the variance of other trait groups. Third, the degree of covariation among trait groups is itself an empirical question that varies across taxa, spatial scales, and ecological contexts (Laughlin 2013, Mouillot et al. 2021). Assuming a priori that all traits converge onto a single functional axis forecloses the

possibility of detecting cases where ecological dimensions are genuinely independent – precisely the cases where a modular approach reveals the most complementary information.

Identifying functionally distinct species (i.e. those with unique trait combinations) might be particularly sensitive to these issues. Because each of these species sustains a portion of the trait space that no other species occupies, their loss cannot be buffered by functional redundancy (Mouillot et al. 2013, Carmona et al. 2021b, Su et al. 2021, Toussaint et al. 2021). Conservation frameworks increasingly recognize the importance of safeguarding these unique ecological strategies since such species often play irreplaceable roles in ecosystems (Carmona et al. 2021a, Mouquet et al. 2025), yet their identification depends critically on how functional space is constructed. Combined trait spaces may dilute or obscure distinctiveness, leading to systematic underestimation of functional rarity and, ultimately, misinformed priorities for conservation.

The trait combination therefore has practical consequences. Functional distances between species in a functional space may reflect, for instance, shared size or reproductive rates, while obscuring their divergence in foraging or habitat use. Metrics calculated from such spaces (e.g. functional richness, uniqueness, divergence) thus risk overgeneralizing or misrepresenting ecosystem processes. This could underestimate the functional richness calculated in a combined trait space, and the community may be wrongly interpreted as functionally homogeneous. Such distortions may ultimately affect our ability to detect the impacts of species loss or invasions and hence design appropriate conservation strategies that preserve ecosystem functioning.

Efforts to disentangle combined functional spaces have been undertaken across several taxa, and the accumulated evidence consistently shows that separating trait dimensions reveals ecologically meaningful patterns. In freshwater fishes, morphological and life-history traits varied largely independently across European river basins, each capturing distinct axes of community assembly that were invisible in combined spaces (Kuczyński et al. 2018). In non-native birds, separating morphological from life-history trait spaces revealed contrasting patterns that would be averaged out in a combined space (Toussaint et al. 2024). Morphological traits describe physical form and biomechanical constraints on movement and foraging, whereas life-history traits describe energy allocation strategies across growth, reproduction, and survival – two dimensions that covary through body size allometry but remain substantially independent (Pigot et al. 2020). When analysed separately, introduced species emerge as morphologically distinct from natives while remaining broadly similar in their life-history strategies. Iconic threatened taxa such as the great Indian bustard *Ardeotis nigriceps* and the Amsterdam albatross *Diomedea amsterdamensis* further illustrate how a species can occupy an extreme position along one functional axis while being functionally ordinary along another. Introduced bird species tend to occupy more peripheral positions in morphological trait space than native species – exhibiting on average larger body mass, longer wings,

and more generalist beak morphologies – while their life-history traits, including clutch size, incubation period, and generation length, fall within the range of native species in recipient communities (Toussaint et al. 2024). This morphological–life history decoupling suggests that invasion success is mediated by locomotor and dispersal capacities rather than by reproductive rate, a pattern that would be undetectable in a combined trait space where morphological and life-history axes are compressed into a single distance gradient.

In marine megafauna, separating trophic from locomotor traits revealed that the species most vulnerable to extinction – large-bodied sharks and rays, such as filter-feeding whale sharks *Rhincodon typus* (uniquely large-bodied, pelagic, planktivorous) or benthic electric rays (family Torpedinidae, uniquely combining benthic lifestyle with electric prey-stunning) – are not only phylogenetically unique but occupy regions of trophic and locomotor space with no functional neighbours, meaning their loss would simultaneously remove ecological functions that no other species can perform.

From an applied perspective, conservation tools that rely on combined functional spaces to identify priority species – such as the FUSE index (Pimiento et al. 2020) or functional vulnerability frameworks (Auber et al. 2022) – may therefore systematically overlook species whose uniqueness is expressed in a single functional dimension, precisely the species whose loss would produce the most idiosyncratic and irreversible functional changes. Disentangling functional spaces is therefore not a technical refinement but a necessary step toward a more faithful and ecologically grounded representation of biodiversity.

These findings call for a shift from a combined functional space toward a more modular approach, where distinct function-related trait spaces are analyzed separately. Understanding how organisms occupy and partition ecological roles thus requires treating these roles as multiple and distinct, each requiring its own trait space and corresponding ecological interpretation. Moreover, such a multidimensional framework would be more explicitly grounded in the processes associated with each function, thereby offering a greater predictive power for guiding conservation decisions.

We here propose to shift from a single functional space aggregating diverse types of traits to multiple function-related trait spaces (Fig. 1). Rather than replacing existing methods, it complements them by allowing researchers to tailor functional spaces to their specific ecological questions. We define function-related trait spaces as distinct multivariate spaces constructed from traits reflecting a specific ecological function or type of traits such as morphology, life-history, or diet. While functional traits often covary due to shared evolutionary and allometric constraints (e.g. body size scaling reproductive traits or trophic levels), this does not preclude the existence of distinct and partially independent axes of ecological variation, each capturing a different aspect of how species acquire resources, move through their environment, and invest in reproduction. Morphology, life-history, and diet represent functionally interpretable domains that reflect different axes of ecological roles – such as dispersal ability,

reproductive strategy, and resource use. Our framework does not assume that these dimensions are completely independent – traits across functional domains inevitably share some covariation due to shared allometric constraints such as body size. However, this covariation is partial, and it is precisely the residual variation that function-related spaces are designed to recover.

To illustrate the value of disentangling combined functional spaces, we applied our framework to the bird clade – one of the best-documented taxa in terms of functional traits – using a global compilation of three types of trait (morphology, life-history, and diet), each capturing a distinct ecological function (Fig. 1). We first tested the extent to which function-related trait spaces describe independent facets of avian ecological roles. We then quantified how each trait space identifies its own set of functionally unique species, leading to potentially divergent conservation priorities depending on the ecological function considered. By comparing rankings across trait spaces, we show that functional uniqueness is not a single, transferable property of species, but a dimension-specific attribute with direct implications for trait-based conservation planning.

Material and methods

Data

We compiled trait information from three databases covering over 10 000 bird species. Morphological traits were obtained from the AVONET database (Tobias et al. 2022), which includes 11 traits for 11 004 species: beak length (culmen and nares), beak width and depth, tarsus length, wing length, Kipps distance, hand–wing index, tail length, secondary length, and body mass. Morphological traits were partitioned according to their primary ecological function following Sayol et al. (2025). Traits associated with locomotion (i.e. tarsus length, wing length, Kipp’s distance, tail length, and body mass) were retained in the locomotion trait space, while beak measurements (i.e. culmen length, beak width, beak depth) were excluded from this space. This distinction is supported by the congruence between the beak-only morphological space and the diet trait space reported by Sayol et al. (2025), and by the well-established functional decoupling between beak morphology and flight or substrate-use traits across birds (Pigot et al. 2020). To test it, we built an independent beak trait space restricted to culmen length, beak width, and beak depth as a validation analysis to assess the congruence between morphological and dietary distinctiveness. Locomotion primarily reflects trophic niche rather than locomotor capacity. The locomotion space therefore captures variation in physical architecture constraining movement and habitat use, independently of dietary adaptation. Reproduction traits were extracted from the AMNIOTE database (Myhrvold et al. 2015), which includes data for 9571 bird species. We selected eight traits: clutch size, number of clutches per year, adult body mass (g), incubation time (days), longevity (years), fledging age (days), egg mass

(g), and body length (cm; tip of beak to cloaca). Diet traits were obtained from EltonTraits (Wilman et al. 2014), which includes data for 9876 bird species. EltonTraits was coded as fuzzy proportions of 10 diet categories per species: invertebrates, vertebrate endotherms, vertebrate ectotherms, fish, unknown vertebrates, scavengers, fruitivores, nectar, seeds, other plant material. Species names across databases were taxonomically harmonized using IUCN taxonomy via the ‘taxize’ R package (www.r-project.org). IUCN Red List categories were retrieved in April 2026 from the IUCN Red List API v4 (‘rredlist’ package, ver. >= 1.0.0) by directly querying every binomial; the most recent assessment was retained for each species. ‘Data Deficient’, ‘Not Evaluated’, ‘Extinct’, ‘Extinct in the Wild’ and ‘Regionally Extinct’ entries were removed from analyses. The final integrated dataset included 8413 species with at least one trait informed from each database. Trait descriptions and sources are listed in the Supporting information.

Missing trait values imputation

Missing values (23% of the trait values were missing; see details in the Supporting information) have been imputed using a machine-learning approach implemented in the ‘missForest’ R package (Stekhoven and Bühlmann 2011). The imputation incorporated morphological, diet, and reproduction traits, as well as phylogenetic information following the protocol of Carmona et al. (2021a). We used the global bird phylogeny from Jetz et al. (2012) defined for 9993 species, accessed through the ‘megatrees’ package (Li 2024), where a single consensus tree was generated from the set of 100 trees. We extracted the first 10 phylogenetic eigenvectors from a PCA of the square-rooted cophenetic distance matrix (Penone et al. 2014) to include phylogenetic signal in the imputation. This way, we included the evolutionary relationships between species in the imputation process by considering the first ten phylogenetic eigenvectors, as recommended by Penone et al. (2014). While phylogenetic diversity is based on the phylogenetic distance between pairs of species, using phylogenetic information in the imputation considers the interaction between all traits and the information, so that the positions imputed in the trait space are much more accurate (Stewart et al. 2023). We demonstrated the stability of our imputation framework considering using different numbers of phylogenetic eigenvectors (i.e. 0, 5, 10, 50, and 100) and multiple imputation (see details in the Supporting information).

Trait spaces

The construction of the trait spaces followed the procedure described by Carmona et al. (2021a). Briefly, quantitative traits were log-transformed and scaled in order to have the same weight between variables. Since all traits were not quantitative (e.g. diet traits were coded as fuzzy traits), we used for all types of traits (i.e. even if all traits were quantitative) a mixed-variables coefficient of distance generalizing Gower’s general coefficient of distance to allow the treatment of various statistical types of variables when calculating

distances. We used the function *ktab.list.df()* in the package ‘ade4’ (Dray and Dufour 2007), which computes a mixed-variable distance coefficient that generalize Gower’s distance by accommodating various types of variables. We used three blocks for each species group (i.e. morphology, life-history, and diet traits) allowing us to mix three ecological types of traits with different numbers of variables and different types of coding. Using this approach allows us to consider each block separately, in particular diet traits, which are coded as fuzzy traits, and then compute a mixed Gower’s distance. The *dist.ktab* function with block structure achieves a form of functional weighting: each block (morphology, life-history, diet) contributes equally to the total distance regardless of the number of traits it contains, which is equivalent to weighting traits by $1 / (\text{number of traits in their block})$.

We then performed a PCoA on the resulting distance matrix and we retained the two first axes, as they capture the highest proportion of total variance and are most commonly examined (Supporting information). By using such a framework, all functional spaces were comparable in subsequent analyses. Notice that using a PCoA on quantitative traits instead of PCA does not affect the position of species within the functional space (Procrustes tests = 1, $p < 0.001$). We compared trait spaces using Procrustes tests to test whether the distribution of species within the different trait spaces was different or not. By doing this, we aimed to detect how the general structure of the trait spaces is affected by the type of trait considered (Supporting information).

Functionally distinct species

We defined functional distinctiveness as an integrative descriptor encompassing two dimensions. Thus, a species is considered functionally distinct when it is both globally isolated from the species pool and locally distinct from its nearest ecological equivalents. Thus, we calculated the functional uniqueness (FUn) which refers to the mean pairwise functional distance from a focal species to all other species in the dataset (Carmona et al. 2019), and functional specialization (FSp), which refers to the distance to the five nearest functional neighbors, capturing local isolation within trait space. These two metrics are complementary – FUn reflects broad-scale functional rarity, FSp reflects fine-scale ecological uniqueness – and their joint consideration provides a more complete characterization of a species’ position in functional space than either metric alone.

Functionally distinct species were identified independently for each trait space. We then compared the top 10% most functionally specialized and unique species across trait spaces to assess consistency. Specifically, we quantified the degree of concordance in species identified as functionally distinct across different trait spaces. We classified each functionally distinct species according to whether it was identified as distinct in only one trait space (i.e. exclusive) or in two (or more) trait spaces (i.e. shared). The proportion of exclusive functionally distinct species provides a measure of how much the identification of functional distinctiveness depends on the choice of trait space. Then we quantified the taxonomic

representation of functionally distinct species, by cross-referencing with family-level classifications. Specifically, we identified the proportion of functionally distinct species that were exclusive to a single trait space (i.e. families that contain species classified as distinct only in locomotion, only in reproduction, only in diet, or only in combined trait space).

To evaluate the robustness of the identification of functionally distinct species across different trait spaces, we conducted a sensitivity analysis based on a resampling framework. We first defined the observed set of rare species in each trait space as the 10% of species with the highest functional uniqueness values. Functional distinctiveness was thus computed from the full set of traits available within each space. We then tested whether the overlap of these functionally distinct species between pairs of trait spaces was robust to the choice and number of traits. For each trait space, we generated 1000 resampled versions by randomly selecting subsets of traits (maintaining the trait type) and recomputing functional distinctiveness. For each resampled trait space, we identified the top 10% most functionally distinct species and quantified their overlap with the corresponding resampled set from another space using the Jaccard index. The distribution of Jaccard overlaps obtained from the resampling provided a null expectation of overlap driven by trait selection alone. For each pair of trait spaces, we compared the observed overlap of functionally distinct species (based on the full trait set) with the distribution of overlaps from the resampling procedure. Observed values lying outside the 95% confidence interval of the resampling distribution were considered evidence that the agreement (or disagreement) between trait spaces was not attributable to random trait selection but reflected consistent cross-space identification of functionally distinct species.

Results

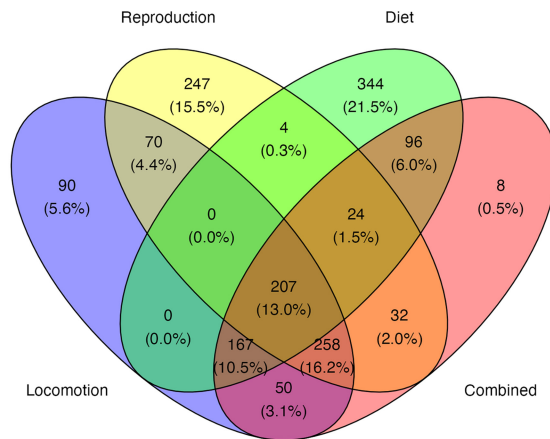
We assembled > 10 000 bird species using three global trait datasets: AVONET for morphology (Tobias et al. 2022), AMNIOTE for life history (Myhrvold et al. 2015), and EltonTraits for diet (Wilman et al. 2014). From these, we built three function-related trait spaces – locomotion, reproduction, and diet – alongside a combined trait space (Supporting information). The locomotion trait space comprises only morphological traits directly associated with locomotion capacity – tarsus length, wing length, Kipp's distance, tail length, and body mass – explicitly excluding beak measurements, which reflect trophic rather than locomotor function (Sayol et al. 2025). To test whether these groupings reflect distinct ecological dimensions, we conducted a confirmatory factor analysis (Supporting information). The model with three latent factors corresponding to locomotion, reproduction, and diet fit the data far better than a single-factor (global) model ($\Delta\chi^2 = 1862.9$, $df = 3$, $p < 0.001$). The modular model also showed lower AIC and BIC scores, while the combined model exhibited poor fit (RMSEA = 0.29), reinforcing the biological relevance of separating traits into function-related spaces.

We used principal coordinates analysis (PCoA) based on a mixed-variable distance coefficient – a generalization of Gower's general similarity coefficient – to accommodate the heterogeneous statistical nature of the trait variables considered (continuous, ordinal, nominal). This approach allowed us to compute pairwise distances between species in a multi-dimensional trait space while preserving the global variance structure of the dataset, such that the resulting ordination axes are ordered by their contribution to total inertia. Both continuous and categorical traits were included in a way that ensures that trait type does not bias the analysis (Supporting information). For locomotion and reproduction, the trait spaces were built on the first two PCoA axes, as they account for more than 70% of the variance. However, for diet traits and the combined trait space, four axes were required to reach 70% of total variance explained (Supporting information). Bootstrap resampling showed that the locomotion and reproduction spaces were barely affected by the number of traits, whereas the diet and combined spaces varied more strongly depending on the number and type of traits included (Supporting information). This sensitivity was particularly pronounced for fuzzy-coded diet variables, highlighting the influence of data type on space structure.

Comparisons between function-related and combined trait spaces revealed striking differences. In the combined space, the dominant axis (combined-axis 1) was almost entirely explained by morphological and life-history traits (i.e. body size and reproductive pace, $R^2 = 0.98$ and 0.79 , respectively). Diet explained far less of this axis ($R^2 = 0.60$). In contrast, combined-axis 2, and subsequent axes, were more correlated to diet traits ($R^2 > 0.75$). These results showed that combination compressed the ecological spectrum into a size-related gradient, while trophic strategies were relegated to secondary axes. Ecologically meaningful patterns, such as the triangular structure of bird trophic niches (from specialist frugivores to strict carnivores), disappear in the combined space.

Our analysis revealed that the way functional space is constructed has critical consequences for detecting functionally distinct species (Fig. 2A). Functional distinctiveness was used as an integrative descriptor encompassing two dimensions: a species is considered functionally distinctive when it is both globally isolated from the species pool (high functionally unique, FUn) and locally distinct from its nearest ecological equivalents (high functionally specialized, FSp). Using the top 10% most FUn and FSp species as a benchmark, we found substantial discordance among trait spaces. For FSp, only 207 species (13%) were consistently identified across all four trait spaces, highlighting the limited overlap when combining locomotion, reproduction, diet, and combined trait information. By contrast, large numbers of species were solely highlighted within individual spaces: 90 species (5.60%) in locomotion, 247 species (15.50%) in reproduction, 344 species (21.50%) in diet, and 8 species (0.50%) in the combined space. For FUn (Fig. 2B), the overlap across all spaces remained limited, with only 45 species (2.10%) consistently shared. Each trait space highlighted a substantial number of unique species: 445 species (20.30%) for locomotion, 393

(a) Functional Specialization



(b) Functional Uniqueness

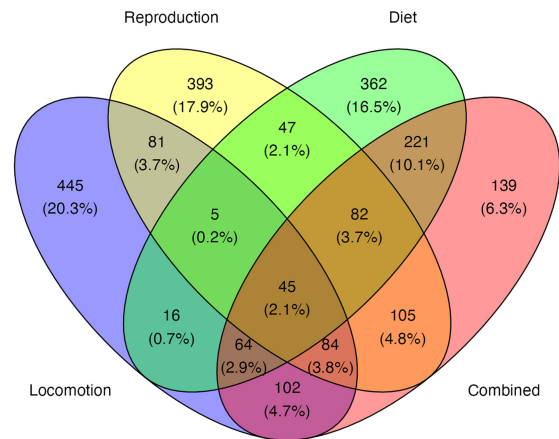


Figure 2. Overlap of the top 10% most functionally specialized and unique bird species across trait spaces. Venn diagrams show the number and percentage of species identified as (a) functionally specialized and (b) functionally unique within the four trait spaces. For each trait space, we considered the top 10% most functionally specialized or unique. Each ellipse represents the set of species classified as highly specialized or unique within a given space, while intersections indicate species simultaneously ranked as such in multiple dimensions.

species (17.90%) for reproduction, 362 species (16.50%) for diet, and 139 species (6.30%) for the combined space.

Family-level analyses further underscored these discrepancies (Fig. 3). In the diet space, Trochilidae (hummingbirds) and Accipitridae (raptors) stood out as highly distinct due to their particular foraging adaptations. In the reproduction space, Anatidae (ducks, geese) and Phasianidae (gamebirds) dominated, reflecting unusual reproductive schedules and clutch sizes. In the locomotion space, Trochilidae again emerged as unique, driven by their hovering flight. The combined space, however, blurred these signals, distributing distinctiveness more evenly across families and obscuring functional specialists. A permutation test confirmed that function-related spaces captured significantly more uniquely distinct species than the combined space (mean proportion of unique species: diet = 0.46, reproduction = 0.30, locomotion = 0.32 versus combined = 0.06; all $p < 0.01$). We tested whether the number of traits, rather than their type, explained discrepancies. A bootstrap procedure varying the number of traits within each space showed that the identity of distinct species remained stable once the minimal structural integrity was met (Procrustes $R > 0.8$; Supporting information). Thus, the differences between the spaces were ecological rather than statistical, each space capturing a distinct aspect of the bird functional strategies.

The conservation status of functionally unique species differed substantially from that of the global bird assemblage (Fig. 4). While ‘Least Concern’ species still represented the majority across all spaces, their dominance declined from 81% in the total pool to 64–72% among unique species. This reduction was mirrored by a relative enrichment of threatened categories, with ‘Vulnerable’, ‘Endangered’, and ‘Critically Endangered’ species collectively accounting for

15–25% of the functionally unique assemblages – nearly twice as their proportion among the global bird fauna. Patterns were particularly pronounced in the locomotion and reproduction spaces, which count the highest proportion of vulnerable species with distinctive ecological strategies. Species unique across all functional dimensions had the fewest ‘Least Concern’ taxa (64%) and the greatest proportion of threatened species (34%), implying that their loss would disproportionately erode the edges of functional trait space.

Discussion

Trait-based ecology has transformed our understanding of biodiversity, but the widespread reliance on combined functional spaces risks oversimplifying ecological complexity. In this study we demonstrated that disentangling functional spaces uncovers ecological variation and conservation priorities that are overlooked by combined approaches. Our global bird analysis showed that different trait groups capture complementary ecological dimensions, and that merging them into a single space obscures meaningful variation. In particular, combined spaces underestimate functional distinctiveness, overlooking hundreds of species that play irreplaceable ecological roles.

The different types of trait differ profoundly in their ecological functions and evolution histories and hence pose the longstanding issue of dimensionality (Laughlin 2013, Carmona et al. 2021a, Mouillot et al. 2021). For birds, life-history traits (e.g. clutch size, lifespan) capture demographic strategies of survival and reproduction. Morphological traits (e.g. wing shape, body shape) are more related to locomotion strategy and habitat-related constraints. Diet traits

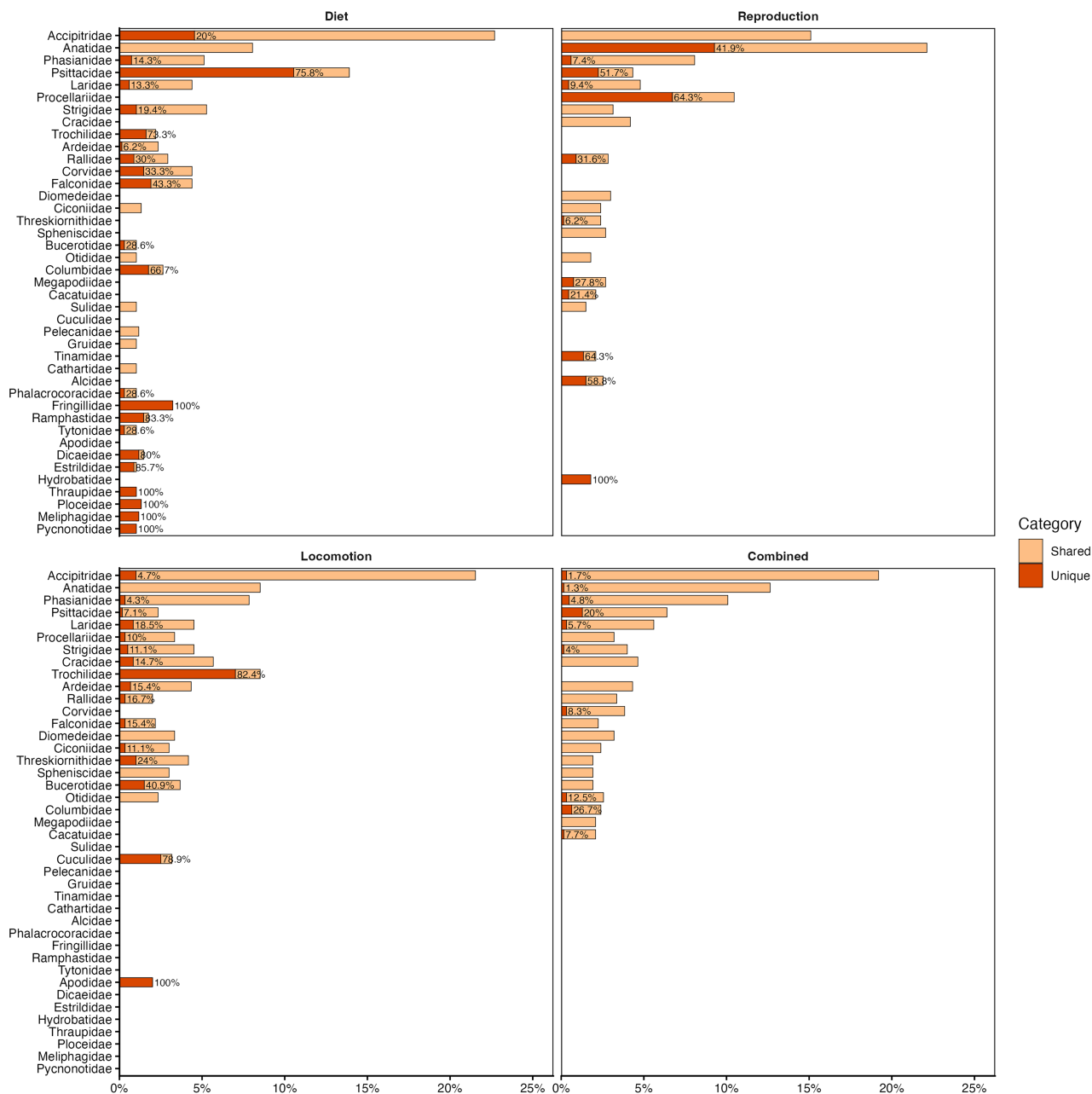


Figure 3. Distribution of the top 10% most functionally distinct bird species across families and functional spaces. Each panel represents the proportion of functionally distinct species within major avian families identified as either unique (dark orange) or shared (light orange) within a given functional space: diet, reproduction, locomotion, and combined. Functionally distinct species were identified as the sum of functionally specialized (FSp) and functionally unique (FUN). For each trait space, we considered the top 10% most functionally specialized or unique. Bar lengths indicate the percentage of species per family falling into each category, with values displayed for the most representative families.

indicate resource use and trophic interactions. We demonstrate that each of these aspects might thus be summarized by distinct ecological axes – often shaped by different selection pressures (Deline et al. 2018, Pigot et al. 2020). Moreover, functional spaces are dominated by traits with strong correlation with others, most commonly body size (Carmona et al. 2021a). This can skew interpretations and create an illusion

of functional completeness. For instance, the two-axes space summarizing bird traits appeared statistically robust but might in practice reflect only variations in size and fecundity, omitting important trophic differences (Carmona et al. 2021a, Bello et al. 2025). We argue that such underestimation of trait-based approaches and functional diversity metrics stems from a lack of clear links between functional traits

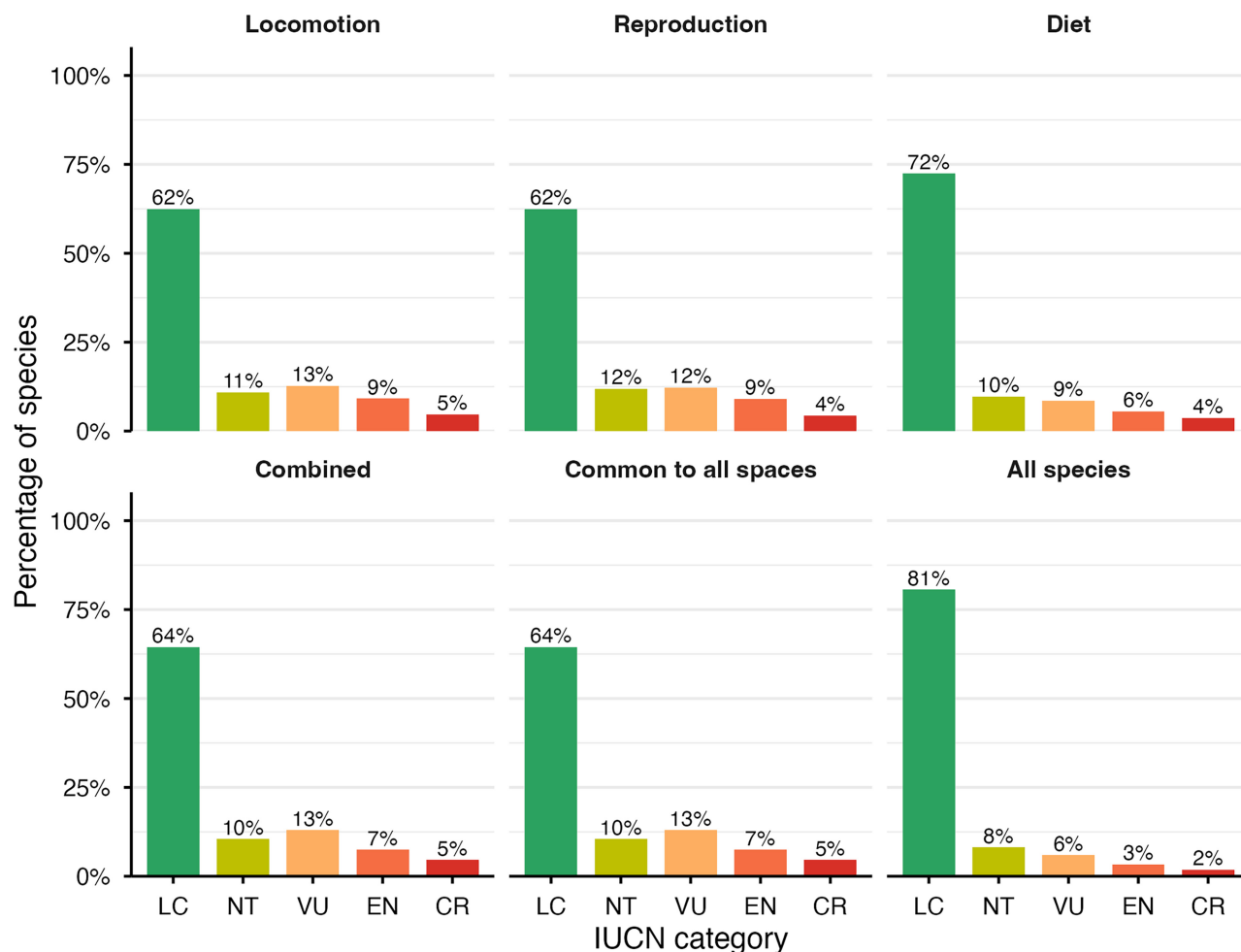


Figure 4. Conservation status of the top 10% most functionally distinct bird species across functional spaces. Bars represent the proportion of IUCN Red List categories (LC=‘Least Concern’, NT=‘Near Threatened’, VU=‘Vulnerable’, EN=‘Endangered’, CR=‘Critically Endangered’) among species identified as functionally unique in each trait space: locomotion, reproduction, diet, combined, and those common to all spaces. For each trait space, we considered the top 10% most functionally specialized or unique. The ‘All species’ panel (bottom right) shows the global distribution of IUCN categories across the entire avian dataset. Percentages indicate the relative proportion of species within each category.

and well-defined ecological processes (Bello et al. 2025) and results in a functional representation that is mathematically valid but ecologically oversimplified.

For conservation, strategies that rely only on combined spaces miss functionally unique species and families that sustain critical ecological functions, from specialized pollinators like hummingbirds to long-lived waterfowl with unique reproductive schedules. By disentangling functional spaces into locomotion, reproduction, and diet, we recover these hidden dimensions and identify distinct conservation priorities. Function-related spaces highlight the particularities of specialized clades, reveal complementary dimensions of functional rarity, and prevent the systematic underestimation of ecological distinctiveness. For instance, mapping species positions across the four functional spaces revealed that functional uniqueness strongly depends on the underlying trait dimensions (Fig. 5). In locomotion space, extreme body forms and locomotion strategies characterized

outliers such as the bearded vulture *Gypaetus barbatus* and common ostrich *Struthio camelus*, while the reproduction space highlighted species with contrasting life-history syndromes, from the long-lived cape shag *Phalacrocorax capensis* to the fast-reproducing Chukar partridge *Alectoris chukar*. The diet space identified distinct trophic specialists such as the Honduran emerald *Amazilia luciae* and blue-winged macaw *Primolius maracana*. When all traits were combined, most species converged toward a central region of the multidimensional space, but a few – including the Japanese sparrowhawk *Tachysiza gularis* and cape shag – remained peripherally positioned, maintaining high functional distinctiveness. These results indicate that functionally unique species differ among trait dimensions, emphasizing that combined functional spaces may conceal key axes of ecological uniqueness.

Our framework does not replace existing approaches, but rather complements them by greater capacity to detect

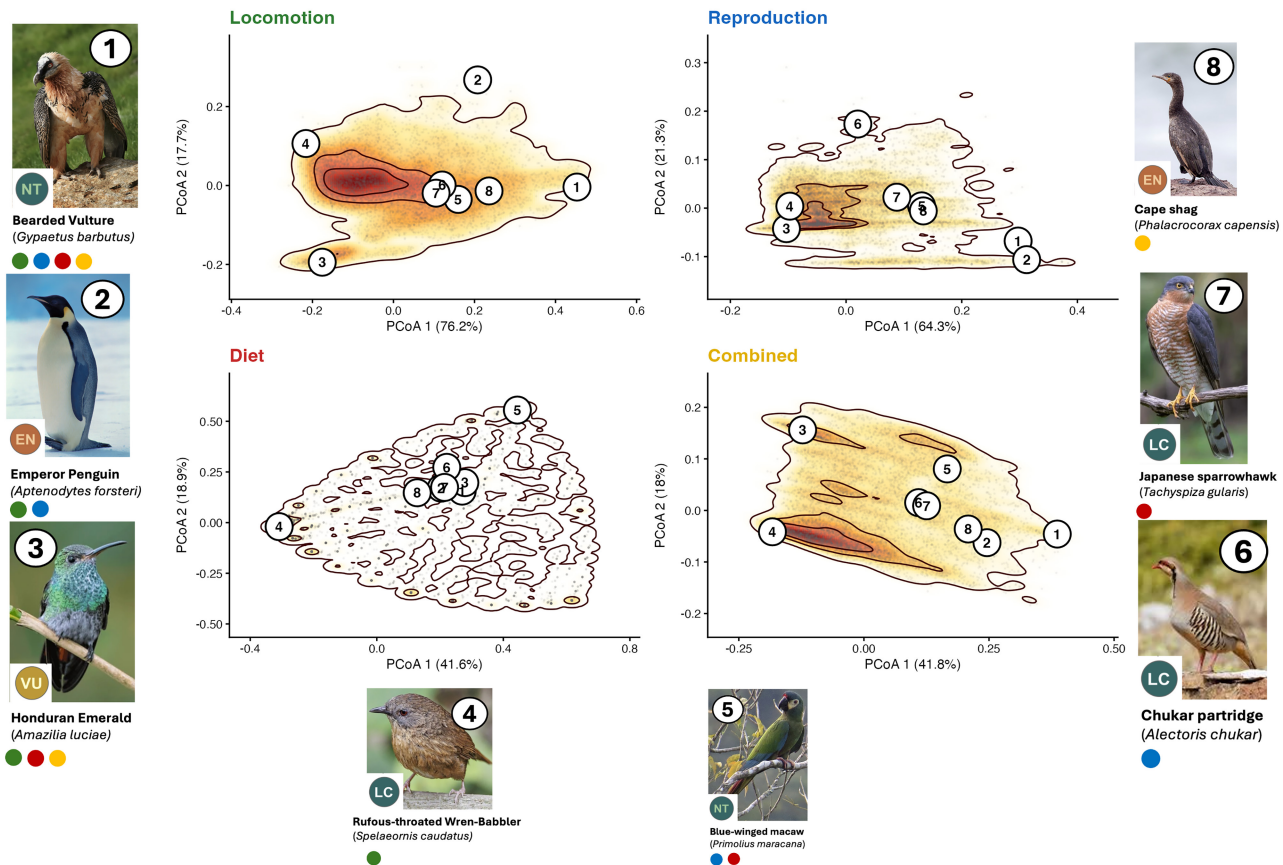


Figure 5. Positions of the top 10% most functionally distinct bird species across four trait spaces. Each panel represents a two-dimensional functional space derived from a principal coordinates analysis (PCoA) based on (a) locomotion, (b) reproductive, (c) dietary, and (d) combined traits. Background color density indicates species concentration in the functional space, with darker tones representing higher species density and pale regions indicating areas of low occupancy. Black contour lines delimit probability density levels (25, 50, 95%). Numbered circles identify bird species that are functionally distinct (high FSp + FUn values) within each space. The icons beside each panel display photographs of the functionally distinct species, together with their IUCN Red List category (LC = 'Least Concern', NT = 'Near Threatened', VU = 'Vulnerable', EN = 'Endangered'). Colored circles under each photo indicate the trait spaces in which the species was identified as functionally distinct.

domain-specific functional distinctiveness that would otherwise be masked by size-dominated axes in a combined trait space. A key strength of the modular framework proposed here is its flexibility. As global trait databases expand to include additional dimensions of species biology – such as coloration, acoustic traits, or behavioral repertoires – new function-related spaces can be incorporated progressively, each anchored to a defined ecological function such as communication, mating, or social interaction. This adaptability allows the framework to evolve in step with data availability, and to be tailored to specific ecological questions rather than being constrained by the traits that happen to be available at a given time. Ultimately, the modular perspective we advocate here is not a fixed framework but a principled approach to preserving ecological information that would otherwise be lost in combined framework.

Beyond its analytical potential, this framework also addresses a pressing conservation gap: that functional biodiversity remains largely absent from current conservation

assessments. Functionally unique species are disproportionately associated with irreversible losses of functional diversity upon extinction (Mouillot et al. 2013), yet their distinctiveness remains largely invisible to current conservation frameworks that rely primarily on abundance- and range-based threat criteria (Pimienta et al. 2020, Auber et al. 2022). Quantifying functional uniqueness across complementary trait spaces, as proposed here, provides a complementary layer of information to extinction risk assessments, enabling the identification of species whose disappearance would result in disproportionate and non-redundant losses of ecological function at the community level (Auber et al. 2022). Prioritizing their conservation is therefore a precautionary strategy to safeguard functional diversity under uncertainty about which ecological functions will prove critical as environments change. Disentangling functional spaces moves conservation science beyond species counts or combined indices, toward protecting the full spectrum of ecological strategies that sustain ecosystems.

Acknowledgements – We thank the anonymous reviewers for their constructive comments, which helped to improve the quality of the manuscript.

Funding – We are grateful to the genotoul bioinformatics platform Toulouse Occitanie (Bioinfo Genotoul, <https://doi.org/10.15545/1.5572369328961167E12>) for providing computing resources.

Conflict of interest – The authors declare no conflict of interest.

Author contributions

Aurèle Toussaint: Conceptualization (lead); Formal analysis (lead); Software (lead); Visualization (lead); Writing – original draft (lead); Writing – review and editing (equal). **Pablo A. Tedesco:** Writing – review and editing (equal). **Gaël Grenouillet:** Writing – review and editing (equal). **Liis Kasari-Toussaint:** Writing – review and editing (equal). **Sébastien Brosse:** Writing – review and editing (equal).

Transparent peer review

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1002/ecog.08578>.

Data availability statement

All R scripts, processed data, and the manuscript associated with this study are available on GitHub at: <https://github.com/FunTraits/DisentFunSpaces>. Data are available from the Zenodo Digital Repository: <https://doi.org/10.5281/zenodo.21487237> (Toussaint et al. 2026).

Supporting information

The Supporting information associated with this article is available with the online version.

References

- Ackerly, D. D. and Cornwell, W. K. 2007. A trait-based approach to community assembly: partitioning of species trait values into within- and among-community components. – *Ecol. Lett.* 10: 135–145.
- Auber, A. et al. 2022. A functional vulnerability framework for biodiversity conservation. – *Nat. Commun.* 13: 4774.
- Bello, F. de et al. 2025. Raunkiaeran shortfalls: challenges and perspectives in trait-based ecology. – *Ecol. Monogr.* 95: e70018.
- Blonder, B., Lamanna, C., Violle, C. and Enquist, B. J. 2014. The n-dimensional hypervolume. – *Global Ecol. Biogeogr.* 23: 595–609.
- Cadotte, M. W. 2017. Functional traits explain ecosystem function through opposing mechanisms. – *Ecol. Lett.* 20: 989–996.
- Carmona, C. P., Bello, F., Mason, N. W. H. and Lepš, J. 2019. Trait probability density (TPD): measuring functional diversity across scales based on TPD with R. – *Ecology* 100: e02876.
- Carmona, C. P., Bueno, C. G., Toussaint, A., Träger, S., Díaz, S., Moora, M., Munson, A. D., Pärtel, M., Zobel, M. and Tamme, R. 2021a. Fine-root traits in the global spectrum of plant form and function. – *Nature* 597: 683–687.
- Carmona, C. P., Tamme, R., Pärtel, M., de Bello, F., Brosse, S., Capdevila, P., González-M., R., González-Suárez, M., Salguero-Gómez, R., Vázquez-Valderrama, M. and Toussaint, A. 2021b. Erosion of global functional diversity across the tree of life. – *Sci. Adv.* 7: eabf2675.
- Deline, B., Greenwood, J. M., Clark, J. W., Puttick, M. N., Peterson, K. J. and Donoghue, P. C. J. 2018. Evolution of metazoan morphological disparity. – *Proc. Natl Acad. Sci. USA* 115: E3688–E3695.
- Díaz, S. et al. 2015. The global spectrum of plant form and function. – *Nature* 529: 167–171.
- Díaz, S. et al. 2019. Pervasive human-driven decline of life on Earth points to the need for transformative change. – *Science* 366, <https://doi.org/10.1126/science.aax3100>.
- Dray, S. and Dufour, A. 2007. The ade4 package: implementing the duality diagram for ecologists. – *J. Stat. Softw.* 22: 1–20.
- Gagic, V., Bartomeus, I., Jonsson, T., Taylor, A., Winqvist, C., Fischer, C., Slade, E. M., Steffan-Dewenter, I., Emmerson, M., Potts, S. G., Tschardt, T., Weisser, W. and Bommarco, R. 2015. Functional identity and diversity of animals predict ecosystem functioning better than species-based indices. – *P. R. Soc. B* 282: 20142620.
- Grime, J. P. 2006. Trait convergence and trait divergence in herbaceous plant communities: mechanisms and consequences. – *J. Veg. Sci.* 17: 255–260.
- Hooper, D. U., Chapin, F. S., Ewel, J. J., Hector, A., Inchausti, P., Lavorel, S., Lawton, J. H., Lodge, D. M., Loreau, M., Naeem, S., Schmid, B., Setälä, H., Symstad, A. J., Vandermeer, J. and Wardle, D. A. 2005. Effects of biodiversity on ecosystem functioning: a consensus of current knowledge. – *Ecol. Monogr.* 75: 3–35.
- Jarzyna, M. A., Quintero, I. and Jetz, W. 2020. Global functional and phylogenetic structure of avian assemblages across elevation and latitude. – *Ecol. Lett.* 24: 196–207.
- Jetz, W., Thomas, G. H., Joy, J. B., Hartmann, K. and Mooers, A. O. 2012. The global diversity of birds in space and time. – *Nature* 491: 444–448.
- Kopf, R. K., Banks, S., Brent, L. J. N., Humphries, P., Jolly, C. J., Lee, P. C., Luiz, O. J., Nimmo, D. and Winemiller, K. O. 2025. Loss of Earth's old, wise, and large animals. – *Science* 387, <https://doi.org/10.1126/science.ado2705>.
- Kuczyński, L., Côte, J., Toussaint, A., Brosse, S., Buisson, L. and Grenouillet, G. 2018. Spatial mismatch in morphological, ecological and phylogenetic diversity, in historical and contemporary European freshwater fish faunas. – *Ecography* 41: 1665–1674.
- Laughlin, D. C. 2013. The intrinsic dimensionality of plant traits and its relevance to community assembly. – *J. Ecol.* 102: 186–193.
- Li, D. 2024. Megatrees: subsets of randomly selected phylogenies from existing mega-phylogenies. – R package, ver. 1.0.1, <https://CRAN.R-project.org/package=megatrees>.
- Mouillot, D., Bellwood, D. R., Baraloto, C., Chave, J., Galzin, R., Harmelin-Vivien, M., Kulbicki, M., Lavergne, S., Lavorel, S., Mouquet, N., Paine, C. E. T., Renaud, J. and Thuiller, W. 2013. Rare species support vulnerable functions in high-diversity ecosystems. – *PLoS Biol.* 11: e1001569.
- Mouillot, D., Loiseau, N., Grenié, M., Algar, A. C., Allegra, M., Cadotte, M. W., Casajus, N., Denelle, P., Guéguen, M., Maire, A., Maitner, B., McGill, B. J., McLean, M., Mouquet, N., Munoz, F., Thuiller, W., Villéger, S., Violle, C. and Auber, A. 2021. The dimensionality and structure of species trait spaces. – *Ecol. Lett.* 24: 1988–2009.
- Mouquet, N., Mahaut, L., Thuiller, W., Auber, A., Casajus, N., Enquist, B. J., Gaüzère, P., Loiseau, N., Mouillot, D., Munoz,

- F., Villéger, S. and Violle, C. 2025. Spatial insurance of distinct ecological functions. – *Ecol. Lett.* 28: e70266.
- Myhrvold, N. P., Baldrige, E., Chan, B., Sivam, D., Freeman, D. L. and Ernest, S. K. M. 2015. An amniote life-history database to perform comparative analyses with birds, mammals, and reptiles. – *Ecology* 96: 3109–3109.
- Naeem, S. and Wright, J. P. 2003. Disentangling biodiversity effects on ecosystem functioning: deriving solutions to a seemingly insurmountable problem. – *Ecol. Lett.* 6: 567–579.
- Penone, C., Davidson, A. D., Shoemaker, K. T., Di Marco, M., Rondinini, C., Brooks, T. M., Young, B. E., Graham, C. H. and Costa, G. C. 2014. Imputation of missing data in life-history trait datasets: which approach performs the best? – *Methods Ecol. Evol.* 5: 961–970.
- Pigot, A. L., Sheard, C., Miller, E. T., Bregman, T. P., Freeman, B. G., Roll, U., Seddon, N., Trisos, C. H., Weeks, B. C. and Tobias, J. A. 2020. Macroevolutionary convergence connects morphological form to ecological function in birds. – *Nat. Ecol. Evol.* 4: 230–239.
- Pimiento, C., Leprieux, F., Silvestro, D., Lefcheck, J. S., Albouy, C., Rasher, D. B., Davis, M., Svenning, J.-C. and Griffin, J. N. 2020. Functional diversity of marine megafauna in the Anthropocene. – *Sci. Adv.* 6: eaay7650.
- Pollock, L. J., Thuiller, W. and Jetz, W. 2017. Large conservation gains possible for global biodiversity facets. – *Nature* 546: 141–144.
- Rüger, N., Condit, R., Dent, D. H., DeWalt, S. J., Hubbell, S. P., Lichstein, J. W., Lopez, O. R., Wirth, C. and Farrior, C. E. 2020. Demographic trade-offs predict tropical forest dynamics. – *Science* 368: 165–168.
- Sala, E. et al. 2021. Protecting the global ocean for biodiversity, food and climate. – *Nature* 592: 397–402.
- Sayol, F., Reijenga, B. R., Tobias, J. A. and Pigot, A. L. 2025. Ecophysical constraints on avian adaptation and diversification. – *Curr. Biol.* 35: 1326–1336.e6.
- Sosiak, C. E. and Barden, P. 2020. Multidimensional trait morphology predicts ecology across ant lineages. – *Funct. Ecol.* 35: 139–152.
- Stekhoven, D. J. and Bühlmann, P. 2011. MissForest – non-parametric missing value imputation for mixed-type data. – *Bioinformatics* 28: 112–118.
- Stewart, K., Carmona, C. P., Clements, C., Venditti, C., Tobias, J. A. and González-Suárez, M. 2023. Functional diversity metrics can perform well with highly incomplete data sets. – *Methods Ecol. Evol.* 14: 2856–2872.
- Su, G., Logez, M., Xu, J., Tao, S., Villéger, S. and Brosse, S. 2021. Human impacts on global freshwater fish biodiversity. – *Science* 371: 835–838.
- Swenson, N. G. 2016. Trait-based plant ecology. – *Ecology* 97: 3556–3558.
- Tobias, J. A. et al. 2022. AVONET: morphological, ecological and geographical data for all birds. – *Ecol. Lett.* 25: 581–597.
- Toussaint, A., Brosse, S., Bueno, C. G., Pärtel, M., Tamme, R. and Carmona, C. P. 2021. Extinction of threatened vertebrates will lead to idiosyncratic changes in functional diversity across the world. – *Nat. Commun.* 12: 1–12.
- Toussaint, A., Pärtel, M. and Carmona, C. P. 2024. Contrasting impacts of non-native and threatened species on morphological, life history, and phylogenetic diversity in bird assemblages. – *Ecol. Lett.* 27: e14373.
- Toussaint, A., Tedesco, P. A., Grenouillet, G., Kasari-Toussaint, L. and Brosse, S. 2026. Data from: Disentangling functional spaces: a multidimensional framework for ecology and conservation. – Zenodo Digital Repository, <https://doi.org/10.5281/zenodo.21487237>.
- Violle, C., Navas, M.-L., Vile, D., Kazakou, E., Fortunel, C., Hummel, I. and Garnier, E. 2007. Let the concept of trait be functional! – *Oikos* 116: 882–892.
- Westoby, M. 2024. Trait-based ecology, trait-free ecology, and in between. – *New Phytol.* 245: 33–39.
- Wilman, H., Belmaker, J., Simpson, J., Rosa, C., Rivadeneira, M. M. and Jetz, W. 2014. EltonTraits 1.0: species-level foraging attributes of the world's birds and mammals. – *Ecology* 95: 2027–2027.